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Differentiable Optics with ∂Lux: I - Deep Calibration of Flat Field and Phase Retrieval with Automatic Differentiation

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Abstract. The sensitivity limits of space telescopes are imposed by uncalibrated errors in the point spread function, background light, and detector sensitivity. These are typically calibrated with specialized wavefront sensor hardware and with flat fields obtained on the ground or with calibration sources, but these leave vulnerabilities to residual time-varying or non-common path aberrations and variations in the detector conditions. It is therefore desirable to infer these from science data alone, facing the prohibitively high dimensional problems of phase retrieval and pixel-level calibration. We introduce a new Python package for physical optics simulation, ∂Lux, which uses the machine learning framework Jax to achieve GPU acceleration and automatic differentiation, and apply this to simulating astronomical imaging. In this first of a series of papers, we show that gradient descent enabled by autodiff can be used to simultaneously perform phase retrieval and calibration of detector sensitivity, scaling efficiently to inferring millions of parameters. This new framework enables high dimensional optimization and inference in data analysis and hardware design in astronomy and beyond, which we explore in subsequent papers in this series.

1 Introduction

At the vanguard of space-based astronomical imaging, photometry, and spectroscopy, imaging precision is limited by systematics introduced by aberrations in the optics and noise processes in the detector. While problems are ubiquitous in astronomy, we are motivated by several core examples. In exoplanet direct imaging, we want to achieve high resolution and high contrast simultaneously with instruments like JWST Coronagraphy,[1] Aperture Masking,[2] or Kernel Phase[3,4] modes. The astrometric mission Toliman[5] aims to measure precise relative positions of the binary stars α Centauri AB to reveal the gravitational influence of unseen planets; this will require micro-arcsecond, micro-pixel astrometric precision. We may also want to perform high precision photometry, whether with dedicated missions like Kepler[6] and TESS,[7] or as ancillary science with Toliman.

In each case, we face serious limitations from an imperfect knowledge of the Pixel Response Function (PRF) – the map of the intra-pixel and inter-pixel variations in detector sensitivity; and also of the Point Spread Function (PSF) – the diffraction-limited pattern by which light from a point source like a star spreads across a detector.

In this series of papers, we present a new software package, ∂Lux, designed for fitting high-dimensional parametrized physical optics models to astronomical data. By using the Python library Jax,[8] we obtain Graphics Processing Unit (GPU) hardware acceleration, as well as automatic differentiation or 'autodiff'[9] features that enable high-dimensional optimization and inference with gradient descent or Hamiltonian Monte Carlo (HMC[10]). In the present paper, we focus on the particular problem of estimating the PRF and PSF simultaneously - i.e. joint flat field calibration and phase retrieval. In Paper II, we will show how autodiff enables improvements to hardware design, directly calculating and optimizing the Fisher information and therefore fundamental figures of merit of an optical system end-to-end. And in Paper III, we will use this for end-to-end deconvolution of high angular resolution imaging with an unknown PSF. We make ∂Lux available as open-source software on GitHub[1], and encourage interested readers to use and contribute to this package as a community resource. This paper has been compiled using ShowYourWork,[11] so that all figures link to the code used to produce them.

1.1 Phase Retrieval

The PSF depends on the successive planes through which light passes from the entrance pupil of the telescope through to the detector. This can be calculated from physical optics: for a simple camera bringing light from the pupil to focus at a detector plane, the PSF is the Fourier transform of the input wavefront, the regime of Fraunhofer diffraction. In the more general Fresnel regime where the detector is slightly out of focus,

[1] github.com/LouisDesdoigts/dLux

there are additional quadratic phase factors before and after this Fourier transform. In more complicated instruments like coronagraphs,[12] the light might pass through multiple re-imaged pupil and focal planes, being operated on in each. In either case, the trouble is that the PSF is distorted by unknown aberrations - distortions in the wavefront, which can be represented as a spatially-varying phase map across the optical plane.

Phase retrieval is the problem of inferring these aberrations from data,[13] which is in general ill-posed:[14] because of the Hermitian symmetry of the Fourier transform, and because we measure intensity and not electric field in optical astronomy, there is a large space of aberrations that would generate the same intensity PSF. Fortunately, this space can be restricted to physically-realistic solutions and readily solved by algorithms such as the Gerchberg-Saxton algorithm,[15] using ideas from compressed sensing,[16] or by machine learning.[17–19] Phase retrieval was memorably performed to infer and correct the serious aberration on the Hubble Space Telescope mirror at launch.[20] Earlier work in the vein of this paper has shown that the phase retrieval problem can be efficiently solved even in the case of detector nonlinearity by taking a forwards model of physical optics, obtaining partial derivatives with autodiff, and optimizing its parameters by gradient descent,[21,22] including a conference presentation of an early version of the present work.[23]

1.2 Detector Calibration

High-precision, high-cadence time series of space photometry from Kepler,[6] K2,[24] TESS[25] and CHEOPS[26] space telescopes have been revolutionary for exoplanetary science[27] and stellar astrophysics.[28,29]

In many practical cases, photometric precision is limited by systematic errors due to variations in sensitivity within and between pixels. Changes in telescope pointing or variations in the PSF with focus couple to these inter- and intra-pixel variations in the flat field to produce changes in overall measured flux. This was particularly severe for the K2 mission where periodic thruster firings introduced a saw-tooth systematic on 6 hour timescales, comparable to the timescales of exoplanet transits and asteroseismic signals of interest.

Much work has been put towards developing data-driven self-calibration of the flat field in K2, which are of general interest and applicability for other space photometry missions. In Self Flat-Fielding (SFF[30]), the systematics are modelled with a spline regression of raw flux versus detector position, together with a linear combination of principal components of all light curves on the detector, and k2sc[31] uses a Gaussian Process (GP[32,33]) similarly. The Causal Pixel Model (CPM[34]) constrains instrumental effects by measuring correlations in the light curves of spatially distant, causally-disconnected pixels. In Pixel Level Decorrelation (PLD[35]) the light curve is detrended by a linear combination of regressors formed by the ensemble of normalized pixel time series contributing to the light curve, together with their higher-order products. Halo photometry[36,37] extracts light curves from a weighted sum of pixels where the weights are learned by minimizing the total variation of the resulting light curve, a convex optimization which is tractable because of autodiff.

Accurate measurement of background noise is also a serious issue for imaging science, where for example bad pixels, background noise, and uncertainties in the detector systematics limit sensitivity of aperture masking interferometry and kernel phase,[38] and for low-surface-brightness science such as with Euclid.[39] In each case it is normal for detectors to be rigorously calibrated on the ground before flight, and with dedicated observations in space; but at the levels of contrast and resolution required for aperture masking on JWST (ideally 10^{-4} at a few λ/D) even small residual miscalibrations can seriously affect performance[2] and we require some method of self-calibration from science data themselves.

2 ∂Lux Differentiable Optical Models

Rather than working with reduced data products like visibilities and light curves, an alternative is to directly model the pixel-level images with a parametrized model of the optics and detector. For example, the popular package poppy[40] can accurately simulate Fresnel and Fraunhofer propagation, with a WebbPSF extension for preset JWST instrument models.[41] In principle, it is possible to fit such a model to data with Markov Chain Monte Carlo (MCMC[42]), fitting the positions of stars and a modal basis representation of aberrations; but common samplers can be very slow, or fail to converge in high dimensions.[43] This restraint is lifted by using HMC, which requires computation of the gradient of the log-likelihood with respect to parameters.

The core technology that enables this for physics also underpins the last decade's revolution in deep learning:[44] autodiff. By application of the chain rule, it is possible to calculate the partial derivatives of

almost arbitrary numerical functions with respect to their floating-point arguments, so long as they are executed in an appropriate software framework. Considerable private-sector and open-source investment has gone into building autodiff frameworks to enable machine learning, such as Theano,[45] TensorFlow,[46] PyTorch,[47] native support in the Julia language,[48] and the framework we apply in this paper, Jax.[8] Jax has an Application Programming Interface (API) mirroring the common Python array library NumPy,[49] while supporting GPU acceleration, Just-In-Time (jit) compilation, and autodiff.

In this paper we introduce a new optical simulation package, ∂Lux, available under an open-source BSD 3 license. It significantly extends the capabilities of morphine,[22,50] a previous Jax optics package from our team based on poppy, being rewritten from the ground up taking full advantage of more-recent Jax features and libraries. It is built in an object-oriented Jax framework called equinox,[51] with an optical system consisting of an equinox module wrapping a stack of 'layers' which are themselves equinox modules applying phase or amplitude screens, or performing Fourier or Fresnel transforms on a wavefront. We use the optimization library optax[52] for gradient-descent optimization, and the probabilistic programming library numpyro[53] for HMC.

There are several other frameworks for automatically-differentiable optics simulation: the most similar is WaveBlocks,[54] a PyTorch package for object-oriented modelling of fluorescence microscopy. The DeepOptics package[55] has been developed in TensorFlow for camera design and computational imaging, and WaveDiff in TensorFlow has been used for detector calibration[56] and PSF modelling.[57] We believe that ∂Lux fills a unique niche in this ecosystem as the most general modelling framework for physical optics. This has driven a design focus on flexibility and ease of open-source contribution and development. The goal is to provide a general object-oriented framework that enables a new user to harness the advantages of autodiff and hardware acceleration provided by Jax with its user-friendly NumPy-like API.

3 Optical and Instrumental Modelling

Within the ∂Lux framework we explicitly define each operation applied to the wavefront as a unique layer. Here we employ a simple optical model with only a pupil and focal plane, defined by 6 sequential layers. Here we show how to create this optical system within the ∂Lux framework:

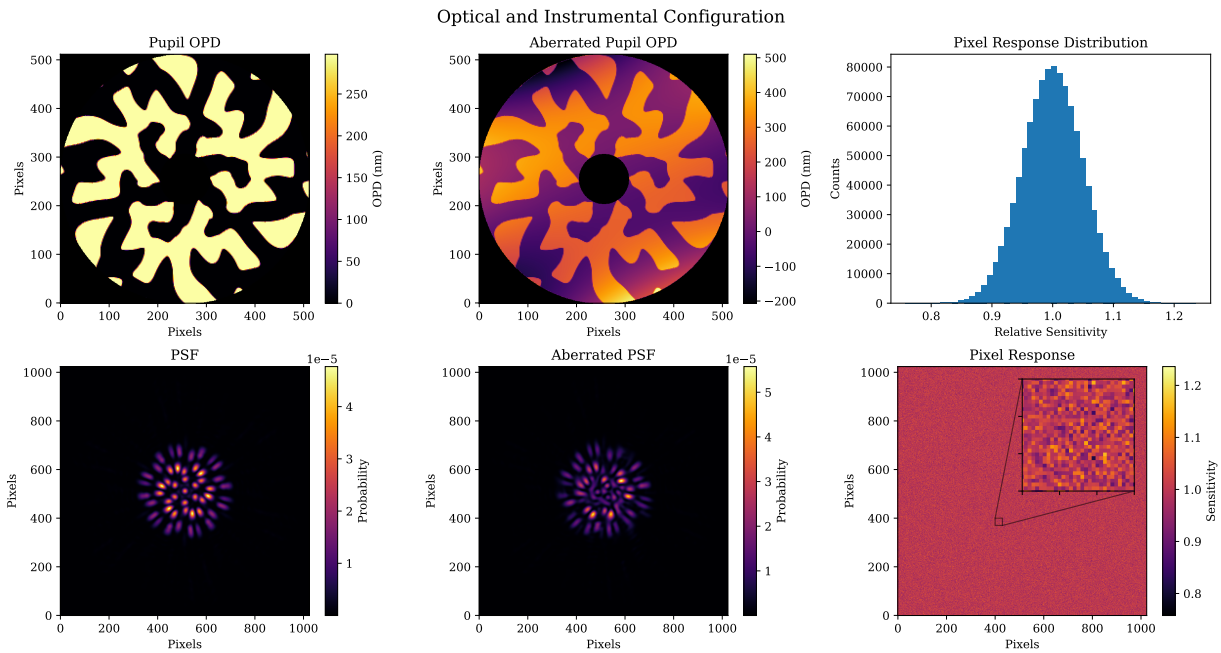

Fig 1 Summary of the optical configuration. Left panel: The top plot show the Optical Path Difference (OPD) of the pupil at the aperture of the telescope. The binary values create a half-wave step at the mean observation wavelength. The bottom plot shows the resulting large single-star PSF without aberrations applied. Middle panel: The top plot shows the total Optical Path Difference (OPD) of the pupil with the optical aberrations applied using low-order Zernike polynomials that is used to generate the data. The bottom plot shows the resulting single-star PSF with the aberrations applied. Clearly these aberrations have a large effect on the PSF and would make recovering information very difficult without appropriate calibration. Right panel: The top plot shows the histogram of the PRF that is applied in the focal plane. These values have a large spread and would greatly affect any results without calibration. The bottom plot shows the full PRF across the whole detector, with a small zoomed region used to show the fine detail that can not be seen when examining the full detector.

```python
import jax.numpy as np
import jax.random as jr
import dLux as dl

# Define wavelengths
wavels = 1e-9 * np.linspace(545, 645, 3)

# Basic Optical Parameters
aperture = 0.5
wf_npix = 512

# Detector Parameters
det_npix = 1024
sampling_rate = 20
det_pixsize = dl.utils.get_pixel_scale(sampling_rate, wavels.mean(), aperture)

# Zernike Basis (ignore piston, tip, tilt)
zern_basis = dl.utils.zernike_basis(10, wf_npix, outside=0.)[3:]
coeffs = 2e-8 * jr.normal(jr.PRNGKey(1), (len(zern_basis),))

# Define Optical Configuration
optical_layers = [
```

```
    dl.CreateWavefront      (wf_npix, aperture),
    dl.CompoundAperture     ([aperture/2], occulter_radii=[aperture/10]),
    dl.ApplyBasisOPD        (zern_basis, coeffs),
    dl.AddOPD               (np.load("mask.npy")),
    dl.NormaliseWavefront   (),
    dl.AngularMFT           (det_npix, det_pixsize)]

# Create Optics object
optics = dl.Optics(optical_layers)
```

The first three layers simply modify the amplitude of the wavefront, representing its response to the opaque components of the optics.

Next the wavefront encounters the binary phase plate diffractive pupil from Toliman,[5] with a half-wave total Optical Path Difference (OPD) between the binary regions. We use this as an information-preserving way to spread the PSF across a large number of pixels which encodes the PRF information for a large number of pixels simultaneously. Alternative methods such as defocusing or using a diffuser[58] come at the cost of eliminating Fourier information which constrain the wavefront and positions. An important consideration which is alleviated by the Toliman pupil is that focal-plane wavefront sensing suffers a sign ambiguity for all phase modes that are inversion-symmetric about the origin, and for unambiguous phase retrieval we therefore need an asymmetric or odd-mode symmetric pupil.[59] This pupil and corresponding aberrated PSF are shown in the left panel of Figure 1.

The wavefront then encounters low-order phase errors parameterised by coefficients on Zernike polynomials. We include the first 4 radial terms, ignoring global piston, tip and tilt for a total of 7 terms. Coefficients were randomly drawn from a normal distribution of mean 0 and standard deviation 20 nm, for a total of 55.8 nm RMS total OPD. These aberrations applied to the pupil mask and resulting PSF are shown in the middle panel of Figure 1, where we see this induces a large amount of distortion in the PSF.

This is then propagated to the 1024×1024 pixel focal plane by a two-sided matrix Fourier transform.[60,61] The detector PRF is modelled by multiplying the resulting PSF by a pixel sensitivity map which is drawn from a normal distribution with mean 0 and standard deviation 0.05. The histogram of pixel sensitivities and total pixel sensitivity map is shown in the right panel of Figure 1. Here we show how to create this within the ∂Lux framework and create an instrument object to model this:

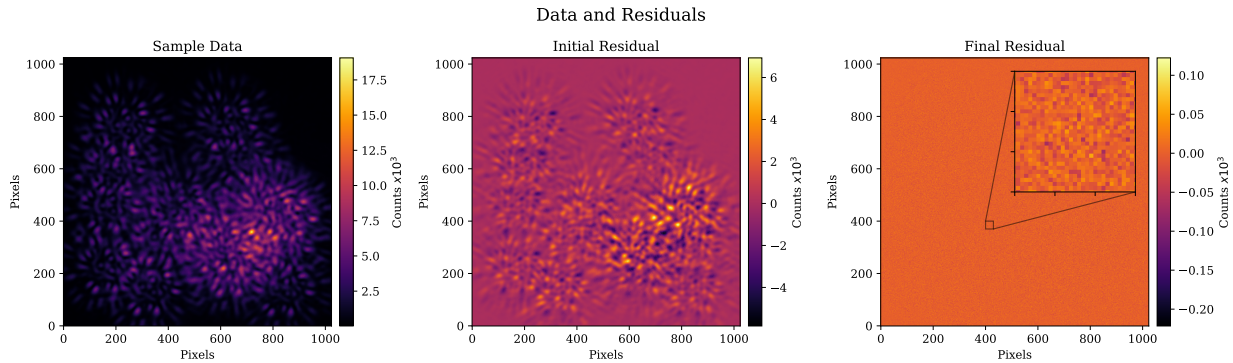

Fig 2 Left: single image of the full four that is used to create the full data set. This image has had both photon and detector noise applied. By eye it is clearly difficult to disentangle the astrophysical information. The large number of overlapping PSFs is chosen in order to spread light across the majority of the detector so that we encode the PRF information for as much of the detector as possible. Middle: the residual of this single image from the data and the initial uncalibrated model. Clearly these residuals are large, showing that there is a large amount of calibration required. Right: this same residual after the model has been optimised. The residual values are much smaller and at the noise floor. A small zoomed region is shown so that the individual pixel-level residuals can be seen.

```
# Pixel response
pix_response = 1 + 0.05*jr.normal(jr.PRNGKey(2), [det_npix, det_npix])

# Create Detector object
detector = dl.Detector([dl.ApplyPixelResponse(pix_response)])

# Combine into instrument
tel = dl.Instrument(optics=optics, detector=detector)
```

This operation is then performed over 15 stars with uniform 100 nm bandpass spectra sampled at three wavelengths 545nm, 595nm and 645nm.. This total model requires the calculation of 30 individual PSFs over a 1024×1024 detector, which takes a total of ~ 1.25 seconds to evaluate on an Apple M1 CPU. The data set is generated by taking this set of images and applying Poisson photon noise to the PSFs and additive Gaussian noise to every pixel as detector read-noise with mean 10 and deviation 2.5 Using this image we recover stellar positions and fluxes, optical aberrations, and individual pixel sensitivities. This gives us 30 stellar positional parameters (15 RA and DEC each), 15 stellar flux parameters, 7 optical aberration parameters, and 1,048,576 detector parameters: the dimensionality of the problem is vastly dominated by the PRF parameters, which is why this problem is made tractable only by using autodiff.

4 Optimisation and Results: Phase Retrieval and Flat Field Estimation

To recover these parameters we initialise a naive unaberrated optical model with a uniform PRF. Stellar positions and fluxes are randomly perturbed from their true values. We will use this model to recover the true values. Figure 2 shows the data and the initial and final residuals.

In order to recover these input aberrations, we perform gradient descent using Adam[62] as implemented in optax,[52] minimizing the posterior of a per-pixel Poisson log-likelihood and a χ^2 log prior on the PRF values, matching the true distribution of mean 1 deviation 0.05. We take the gradient of this loss function with respect to the position and flux of all stars, the response in all pixels, and the Zernike coefficients. We also use a staged optimisation strategy, initially optimising the positions, fluxes and optical aberrations in order to recover a good PSF model before learning the PRF. This is necessary to circumvent the high degree of covariance that each individual pixel experiences with the parameters of the PSF. To achieve this the learning rate of the PRF is set to zero for the first 50 epochs of optimisation, after which the PRF learning rate is initialised while the position, flux and aberrations parameters are frozen for 25 epochs before being allowed to vary again.

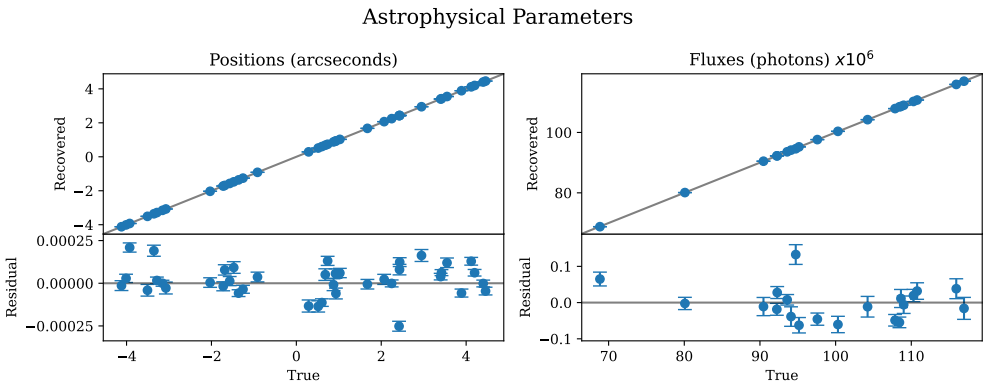

Fig 3 Left: recovery of the individual RA-Dec positions of each star, with the top section showing the correlation between the true and recovered values in units of arcseconds. The bottom section shows the individual residuals for each. Clearly these parameters have been well recovered through the optimisation. Right: the recovery of the flux parameters in units of photons. The top section shows the correlation between the true and recovered values, and the bottom showing the resulting residuals. The error bars are the 1-σ deviations calculated from the covariance Hessian matrix marginalised over positions, fluxes and aberrations.

The optimisation is performed for a total of 100 epochs, with each value-and-gradient calculation taking ~ 2.75 seconds to evaluate on an Apple M1 Max CPU, for a total optimisation time of ~ 4.5 minutes.

This optimisation strategy performs well, with all of the parameters being recovered down to the noise level. This can be seen in Figure 2 with the right panel showing a sample residual of the data and model after optimisation, which is clearly only noise. Correlations and residual plots of the positions and fluxes after the optimisation are shown in Figure 3. The 1-σ error bars are generated by autodiff, calculating the covariance matrix under the Laplace approximation as the negative matrix inverse of the Hessian of the likelihood function evaluated at the maximum likelihood estimate. These errorbars are clearly an underestimate: this is understandable as they are marginalised over positions, fluxes and aberration, but not the PRF residuals which are the dominant noise source. The million-dimensional Hessian required to estimate these uncertainties is at present intractable even in Jax. These parameters are recovered to the expected distribution when performed without modelling the PRF. Figure 4 shows the recovery of the optical aberrations, with a final residual aberration of 0.0499 nm RMS. Finally, Figure 5 shows a correlation plot of the true and recovered sensitivity values and the corresponding residual histogram. The points of the correlation plot are colour-coded with the total counts for each pixel, showing as expected that pixels with higher photon counts having a better recovery of the PRF since these pixels have a greater signal-to-noise ratio (SNR). Overall the parameters are very well recovered down to the noise level.

The optical propagation algorithm employs two-sided matrix Fourier transforms,[60] which can be greatly improved if deployed onto GPU hardware. In our a test case, we initialise from a very naive state far from the truth, requiring a larger number of optimisation epochs than would be required in a real-world example. Furthermore as a calibration algorithm across a whole detector the complexity of the model is greatly increased, applying this to real data would likely be focused on small detector regions resulting in less computationally intensive models.

The use of a forwards model clearly requires a good match between the true and model PSF. The benefit of using differentiable optical models however is that it provides the perfect platform upon which to directly learn this model from the data, with the ability to scale the complexity as required for each problem. Similarly since we are working with direct image data as opposed to reduced data-products, miss-calibrations of the system are simple to diagnose through direct examination of the residuals of the data and model. Figure 2 demonstrates that in this example we can be sure of a good fit because the final residual is pure noise.

This algorithm demonstrates promising performance applied to complex high-dimensional optimisation problems as shown in Section 4. However, this method does have its limitations. To test these limits we fit a single star across a wide range of fluxes and look at the recovered positions and fluxes, following the same optimisation strategy as described in Section 4. Figure 6 shows the recovered positional residuals and fractional flux errors. The error bars on the fluxes are 1-σ calculated as the square-root of the total

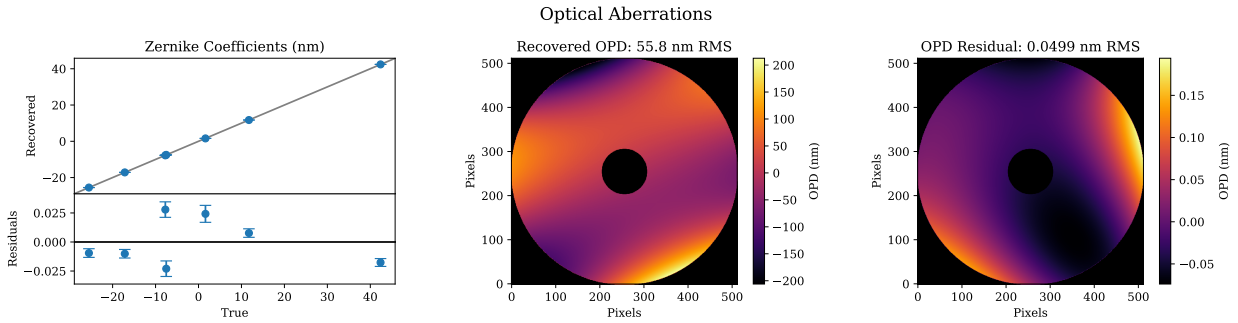

Fig 4 Recovery of the optical aberrations after optimisation. Left: the amplitudes of the Zernike polynomial coefficients, with the top section showing the correlation plot of the true and recovered values, and the bottom showing the individual residuals. Middle and right: the true total OPD and the residuals generated by these values respectively. The error bars are the 1-σ deviations calculated from the covariance Hessian matrix marginalised over positions, fluxes and aberrations.

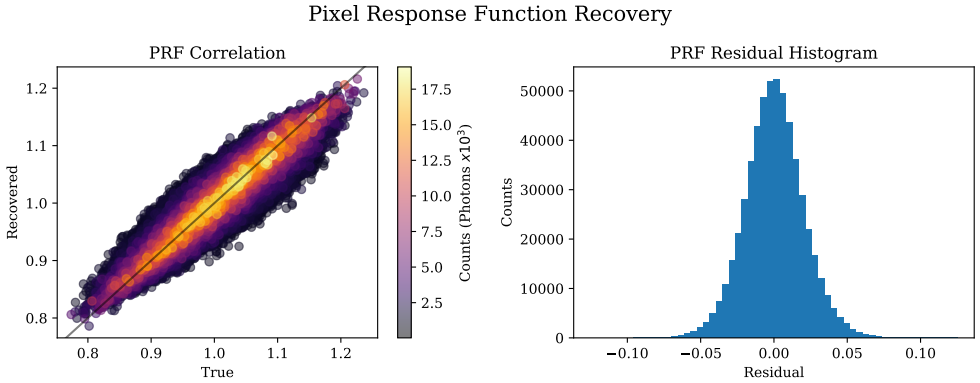

Fig 5 This figure shows the PRF values after optimisation. Left: Correlation between the pixel-level true and recovered values. Each point is color-coded with the total flux incident to each pixel, ie its SNR. The pixels with a greater signal are recovered better as expected. Right: Histogram of these residuals, with the majority of values being very well recovered and small symmetric about a residual of zero.

flux. Clearly using this optical architecture requires more than 10^4 photons in order to have well recovered parameters.

5 Conclusions & Future Work

In this first paper of a planned series, we have demonstrated photometry, phase retrieval, and detector calibration using autodiff to make tractable gradient descent in very high dimensions. Our software ∂Lux integrates with a rich and user-friendly ecosystem of Jax packages for optimization and sampling, which point to a number of natural extensions.

The most obvious is to produce calibrated photometric time series, making use of correlations in time rather than the 'bag of images' approach taken in this work. Instead of independently optimizing each frame, they could be treated hierarchically, with regularization imposed on each star's light curve. GP models are widely used in astronomy, and Jax packages such as tinygp[33] and GPJax[63] permit differentiable optimization and sampling that integrate efficiently with ∂Lux.

Photometry of crowded fields is often of poor quality even from space telescopes, and there has been considerable investment in algorithms for deblending light curves in the K2 galactic plane campaigns[64] and in TESS with its large pixels.[65–67] High dimensional optimization with autodiff in principle renders deblending tractable even close to the diffraction limit with unknown PSFs. Prior knowledge of a structured PSF, such as with the diffractive pupil architecture of Toliman,[68,69] may further improve this deblending and retrieval.

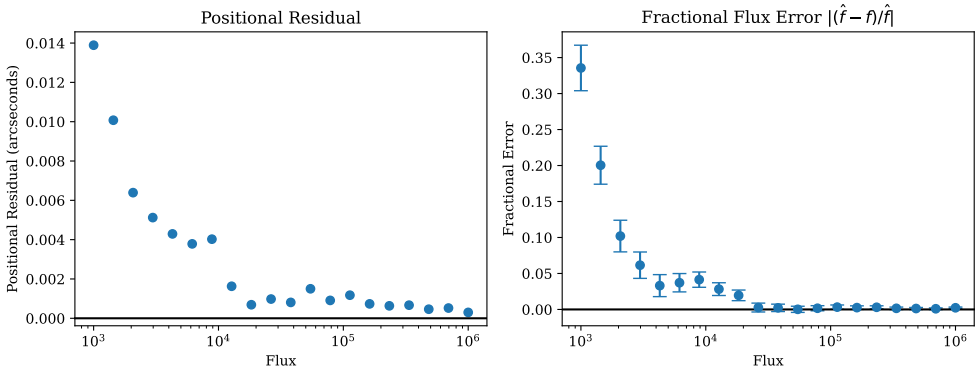

Fig 6 Parameter recovery of a single star over varying SNR. Left: Absolute positional residuals vs total flux. Right: Fractional flux error vs total flux.

High contrast imaging with the recently-launched JWST[70] and the upcoming Roman Space Telescope coronagraph[71] will in each case demand a new generation of data-driven PSF and detector calibrations beyond those used fruitfully on ground-based data.[72] While we have not attempted this in the present work, ∂Lux can flexibly model aberrations in the multiple intermediate planes of these coronagraphs, and in future work we intend to assess the practical and fundamental limits of multiplane phase retrieval and calibration to improve coronagraphic modelling. Uncertainties in the flat field are also the leading contributor to noise in the exoplanet transmission spectroscopy with JWST,[73] and a promising future direction with ∂Lux will be to build models not just of imaging data, but of spectrophotometry.

In Paper II in this series, we will tackle the isomorphic problem of hardware design. While we have tackled phase design in previous work,[22] autodiff permits calculation and optimization of the Fisher information directly,[74] which means we can optimize towards the fundamental limits of parameter sensitivity. In subsequent papers, we intend to explore parametric and nonparametric deconvolution of images at high angular resolution and contrast.

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This research made use of NumPy;[49] Matplotlib;[75] Jax;[8] equinox;[51] and optax.[52]

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